

Systemic Risk and Interbank Connectedness Under Financial Stress

**STAT GR5291: Advanced Data Analysis
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I. Introduction

The stability of the financial system depends heavily on the degree of interconnectedness among financial institutions. During periods of financial stress, shocks originating from one institution or market can spread rapidly to others, increasing the risk of systemic instability. As a result, understanding how financial stress influences connectedness within the banking sector has become an important topic in financial economics and risk management literature.

Previous studies have shown that financial connectedness tends to increase during crisis periods. Francis X. Diebold and Kamil Yilmaz (2014) develop a connectedness framework based on variance decompositions to measure how shocks spill over across financial institutions. Similarly, Geert Bekaert, Campbell R. Harvey, and Angela Ng (2005) show that correlations across financial markets rise significantly during crisis periods due to stronger market integration and higher volatility. In addition, Tobias Adrian and Markus K. Brunnermeier (2016) introduce the CoVaR framework to measure the systemic risk contribution of individual institutions and demonstrate how financial distress can propagate throughout the financial system.

More recent research further emphasizes the importance of dynamic interconnectedness in the banking sector. Kapinos, Kishor, and Ma (2022) find that bank comovement increases during periods of financial strain and macroeconomic uncertainty, suggesting that financial stress strengthens linkages among institutions. Likewise, Patrick McLemore, Atanas Mihov, and Luis Sanz (2022) show that exposure to highly connected global financial markets increases the systemic risk contribution of U.S. banks.

Motivated by this literature, our project investigates how financial stress affects interconnectedness among major U.S. banks. Using bank equity return data, we analyze pairwise return correlations and examine how these correlations change during periods of elevated market stress. In particular, we compare connectedness measures across high-stress and low-stress environments using indicators such as the VIX and the Federal Reserve Bank of St. Louis Financial Stress Index. By doing so, our study aims to provide additional empirical evidence on how financial stress amplifies linkages within the banking sector and increases the potential for systemic risk transmission.

II. EDA

Data Description

This study uses daily financial market data for nine major U.S. financial institutions (AIG, BAC, C, GS, JPM, MS, PNC, USB, and WFC), selected based on their systemic importance. The dataset covers the period from January 2010 to March 2020.

Daily log returns are calculated from adjusted closing prices obtained from Yahoo Finance. Market conditions are captured using the S&P 500 index (MKT) as a proxy for overall market movements. Financial stress is measured using two indicators: the CBOE Volatility Index (VIX) and the St. Louis Fed Financial Stress Index (STLFSI), both obtained from the Federal Reserve Economic Data (FRED) database.

To measure interbank connectedness, we construct the average pairwise correlation among bank returns (AvgCorr). In addition, a binary stress indicator is defined as 1 when the VIX exceeds its 90th percentile and 0 otherwise.

Research Objective

This study examines how financial stress affects interbank connectedness among major U.S. banks. In particular, we analyze whether higher levels of financial stress are associated with stronger return comovement and increased systemic risk.

We further compare interbank relationships across stress and calm periods and explore how connectedness evolves over time. These analyses provide empirical evidence on how systemic risk intensifies under stressed market conditions.

Descriptive Statistics

	count	mean	std	min	25%	50%	75%	max	ann_vol
AIG	2472.0	0.000114	0.020406	-0.233761	-0.008083	0.000516	0.008687	0.151307	0.323934
BAC	2472.0	0.000210	0.021256	-0.227129	-0.009630	0.000358	0.010555	0.163786	0.337422
C	2472.0	0.000249	0.020550	-0.179336	-0.008788	0.000407	0.009744	0.165381	0.326221
GS	2472.0	0.000067	0.017281	-0.136864	-0.008321	0.000445	0.009056	0.161951	0.274329
JPM	2472.0	0.000516	0.016659	-0.145552	-0.007245	0.000511	0.008656	0.165620	0.264451
MS	2472.0	0.000220	0.021561	-0.163460	-0.009824	0.000349	0.011171	0.180403	0.342264
PNC	2472.0	0.000421	0.015688	-0.145622	-0.006755	0.000718	0.008478	0.121717	0.249037
USB	2472.0	0.000354	0.014444	-0.155959	-0.005929	0.000770	0.007328	0.160144	0.229295
WFC	2472.0	0.000195	0.016297	-0.172779	-0.007413	0.000283	0.008125	0.127215	0.258710

Table 1 reports descriptive statistics for daily log returns of the nine major U.S. banks over the sample period from January 2010 to March 2020. Across institutions, average daily returns are close to zero, which is typical for high-frequency financial return data. However, return volatility varies substantially across banks.

When annualized, return volatility ranges from approximately 23% to 34%, indicating meaningful heterogeneity in risk exposure among large financial institutions. Banks such as BAC and MS exhibit relatively higher volatility, while institutions like USB and PNC display lower overall risk.

	Mean (All)	Mean (Stress)	Mean (Calm)	Std (All)	Std (Stress)	Std (Calm)
AIG	0.000114	-0.001168	0.000257	0.020406	0.039859	0.016915
BAC	0.000210	-0.003553	0.000630	0.021256	0.043556	0.017019
C	0.000249	-0.002225	0.000525	0.020550	0.041905	0.016539
GS	0.000067	-0.001577	0.000250	0.017281	0.031637	0.014846
JPM	0.000516	-0.001005	0.000685	0.016659	0.033697	0.013492
MS	0.000220	-0.001962	0.000463	0.021561	0.042966	0.017635
PNC	0.000421	-0.001140	0.000595	0.015688	0.031367	0.012804
USB	0.000354	-0.001382	0.000548	0.014444	0.031375	0.011053
WFC	0.000195	-0.001694	0.000406	0.016297	0.033356	0.013083

To examine how financial stress affects bank performance, we compare descriptive statistics between stress and calm periods, where stress periods are defined as days when the VIX exceeds its 90th percentile. The results show a clear contrast between the two regimes.

During stress periods, average bank returns are generally lower and often negative, while return volatility increases sharply across all institutions. In contrast, calm periods are characterized by higher average returns and substantially lower volatility. This pattern is consistent with the notion that heightened financial stress amplifies uncertainty and downside risk in the banking sector.

```
mean_corr_all, mean_corr_stress, mean_corr_calm

(np.float64(0.7493092680852906),
 np.float64(0.855534250622939),
 np.float64(0.6785992528710659))
```

As a simple proxy for systemic risk, we compute the average pairwise correlation among bank returns. The results indicate that interbank correlations are significantly higher during stress periods than during calm periods. Specifically, average correlations increase from approximately 0.68 in calm periods to 0.86 during stress periods.

This increase in comovement suggests that banks become more tightly connected when financial stress intensifies, raising the potential for risk spillovers and systemic instability.

Visualization

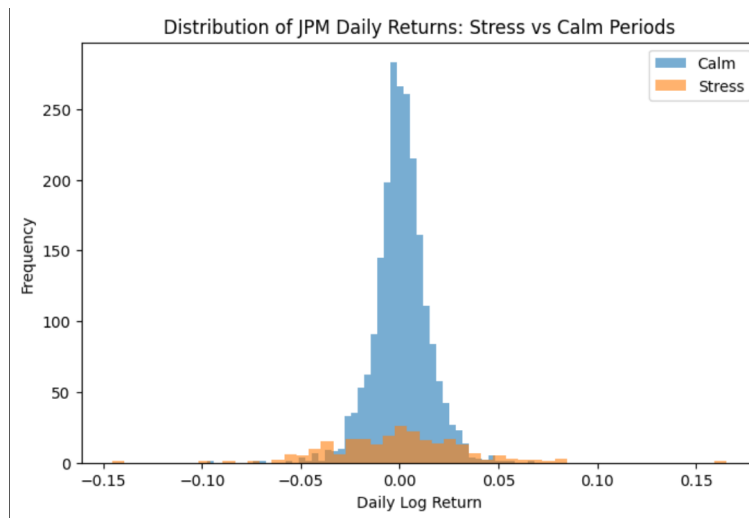


Figure 1. Distribution of JPM Daily Returns: Stress vs Calm

This figure compares the distribution of JPM's daily log returns during stress and calm periods. Returns during stress periods exhibit a much wider dispersion and heavier tails, indicating substantially higher volatility. In contrast, calm periods are characterized by a tighter distribution centered close to zero, reflecting more stable market conditions.

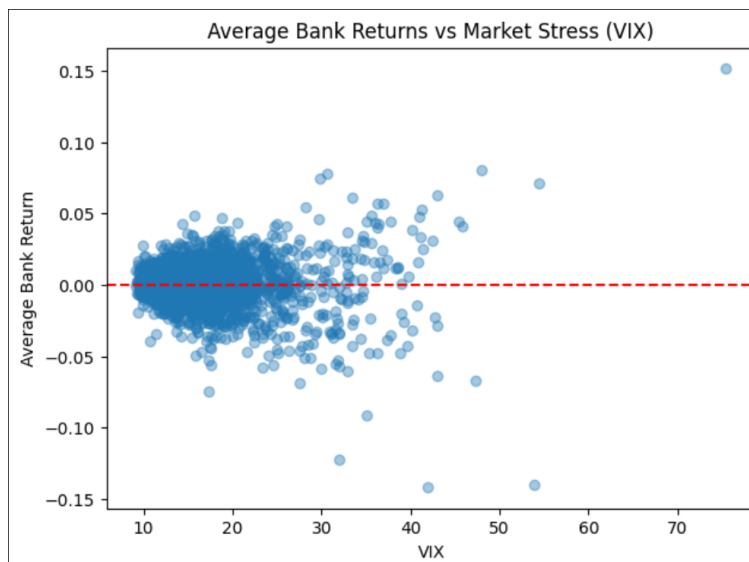


Figure 2. Average Bank Returns vs Market Stress (VIX)

This scatter plot illustrates the relationship between average bank returns and market stress, as measured by the VIX. Higher VIX levels are generally associated with lower and more volatile

average bank returns, with a greater concentration of negative outcomes. This pattern suggests that elevated market stress is linked to weaker performance in the banking sector.

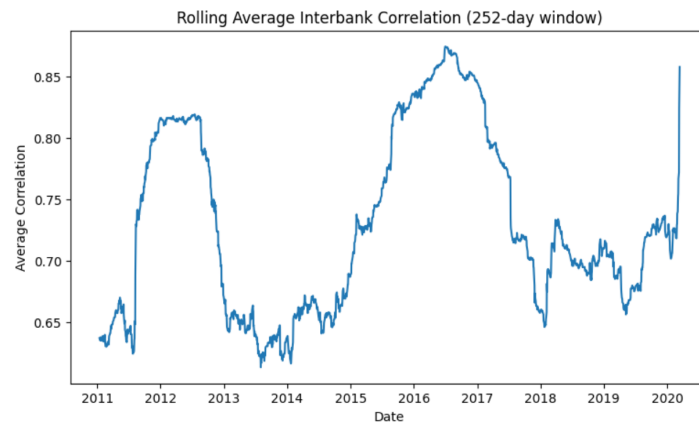


Figure 3. Rolling Average Interbank Correlation (252-day window)

This figure shows the evolution of average interbank correlations over time using a 252-day rolling window. Periods of heightened financial stress are associated with pronounced increases in correlation, indicating stronger comovement among bank returns. These spikes in connectedness highlight the dynamic nature of systemic risk in the banking sector.

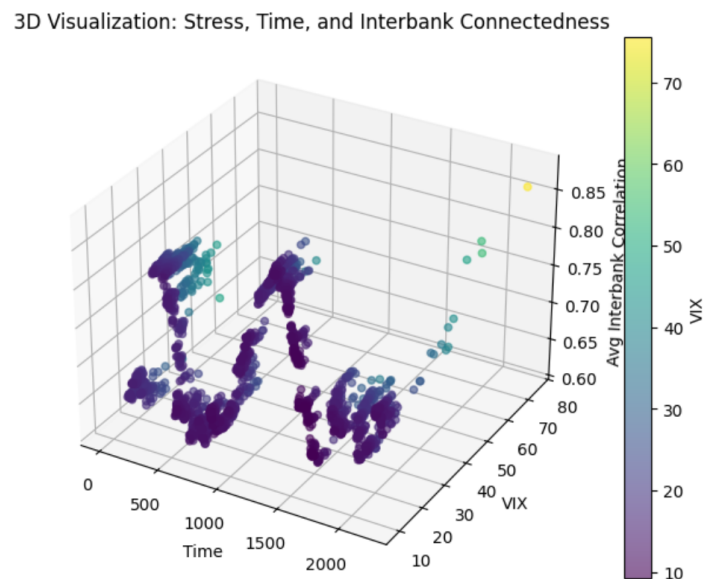


Figure 4. 3D Visualization of Stress, Time, and Interbank Connectedness

This three-dimensional visualization jointly captures the relationship between time, market stress (VIX), and average interbank correlation. Higher levels of market stress tend to coincide with elevated interbank connectedness, particularly during turbulent periods. The plot illustrates

how systemic risk intensifies when financial stress rises, reinforcing the link between stress and network-level dependence.

EDA summary

Overall, the exploratory analysis provides consistent evidence that financial stress is associated with lower returns, higher volatility, and stronger interbank connectedness. These findings suggest that systemic risk intensifies during stress periods.

The observed patterns motivate the regression analysis in the next section to formally quantify these relationships.

III. Model assumptions

Introduction

Before proceeding with regression analysis, we examine whether the standard OLS assumptions are satisfied. In particular, we test for heteroskedasticity, serial correlation, and normality of residuals. Violations of these assumptions may lead to biased standard errors and unreliable statistical inference.

Heteroskedasticity (BP test)

BP test p-value: $4.37e-76$

We first test for heteroskedasticity using the Breusch–Pagan test. The test yields a p-value close to zero, indicating strong evidence against the null hypothesis of homoskedasticity. This suggests that the variance of the residuals is not constant, violating a key OLS assumption.

Autocorrelation

Durbin-Watson ≈ 0.004

We next examine serial correlation using the Durbin–Watson statistic. The value is close to zero, indicating strong positive autocorrelation in the residuals.

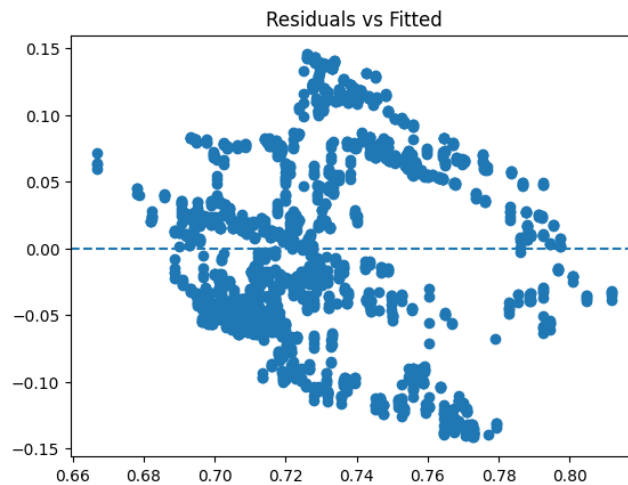
This suggests that the errors are highly persistent over time, violating the independence assumption of OLS.

Normality

JB p-value ≈ 0

We also assess the normality of residuals using the Jarque–Bera test. The test rejects the null hypothesis of normality, indicating that residuals deviate from a normal distribution.

Residual Plot



The residual plot further shows patterns in the variance, supporting the presence of heteroskedasticity.

Conclusion

Overall, the diagnostic tests reveal clear violations of key OLS assumptions, including heteroskedasticity and strong serial correlation. As a result, standard OLS inference may not be reliable. To address these issues, we use heteroskedasticity and autocorrelation consistent (HAC) standard errors in subsequent analysis. In addition, the presence of strong persistence motivates the use of dynamic models that incorporate lagged variables.

IV. Models

Model 1: Baseline Model

We begin with a baseline regression model:

$$\text{AvgCorr} = \beta_0 + \beta_1 \text{Stress} + \varepsilon$$

OLS Regression Results						
=====						
Dep. Variable:	AvgCorr	R-squared:	0.127			
Model:	OLS	Adj. R-squared:	0.127			
Method:	Least Squares	F-statistic:	21.78			
Date:	Sun, 05 Apr 2026	Prob (F-statistic):	3.25e-06			
Time:	05:45:40	Log-Likelihood:	2884.9			
No. Observations:	2220	AIC:	-5606.			
Df Residuals:	2218	BIC:	-5594.			
Df Model:	1					
Covariance Type:	HAC					
=====						
	coef	std err	z	P> z	[0.025	0.975]
const	0.7297	0.007	110.773	0.000	0.717	0.743
Stress	0.0261	0.006	4.667	0.000	0.015	0.037
=====						
Omnibus:	155.831	Durbin-Watson:	0.003			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	58.098			
Skew:	0.118	Prob(JB):	2.42e-13			
Kurtosis:	2.243	Cond. No.	1.00			

The coefficient on Stress is positive and statistically significant, indicating that interbank connectedness increases during stress periods.

Model 2:

We extend the baseline model by incorporating a continuous measure of financial stress:

$$\text{AvgCorr} = \beta_0 + \beta_1 \text{Stress} + \beta_2 \text{STLFSI} + \varepsilon$$

OLS Regression Results						
=====						
Dep. Variable:	AvgCorr	R-squared:	0.135			
Model:	OLS	Adj. R-squared:	0.134			
Method:	Least Squares	F-statistic:	115.7			
Date:	Thu, 23 Apr 2026	Prob (F-statistic):	1.30e-69			
Time:	23:24:39	Log-Likelihood:	2815.8			
No. Observations:	2220	AIC:	-5624.			
Df Residuals:	2216	BIC:	-5601.			
Df Model:	3					
Covariance Type:	nonrobust					
=====						
	coef	std err	t	P> t	[0.025	0.975]

const	0.8317	0.006	130.815	0.000	0.819	0.844
STLFSI	0.0991	0.006	16.987	0.000	0.088	0.111
stress	-0.0483	0.011	-4.581	0.000	-0.069	-0.028
interaction	-0.0623	0.015	-4.249	0.000	-0.091	-0.034

Omnibus:	86.255	Durbin-Watson:	0.004			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	37.973			
Skew:	0.011	Prob(JB):	5.68e-09			
Kurtosis:	2.360	Cond. No.	17.8			

The inclusion of a continuous stress measure allows us to capture variation in the intensity of financial stress beyond the binary indicator. The results show that financial stress remains an important factor in explaining interbank connectedness, with a modest improvement in model fit.

Exploratory Extensions

We also explore additional model specifications to capture potential nonlinear effects. In particular, we estimate an interaction model between financial stress and the stress indicator. While the interaction terms are statistically significant, the model does not resolve the issues identified in the diagnostic tests, such as heteroskedasticity and serial correlation. Therefore, the interaction model does not provide a substantial improvement over the baseline specifications. We further test whether different financial stress measures have equivalent effects by testing the hypothesis $\beta_1 = \beta_2$. The hypothesis is rejected ($p < 0.001$), indicating that different stress measures capture distinct aspects of market conditions.

```
model_scaled.t_test("VIXCLS = STLFSI")

<class 'statsmodels.stats.contrast.ContrastResults'>
      Test for Constraints
=====
      coef      std err          t      P>|t|      [0.025      0.975]
-----
c0         -0.0423       0.004     -11.909      0.000      -0.049      -0.035
=====
```

Model 3: Dynamic Model

Given the strong serial correlation observed in the residual diagnostics, we introduce a lagged dependent variable to capture persistence in interbank connectedness:

$$\text{AvgCorr} = \beta_0 + \beta_1 \text{Stress} + \beta_2 \text{STLFSI} + \beta_3 \text{LagCorr} + \epsilon$$

OLS Regression Results						
=====						
Dep. Variable:	AvgCorr	R-squared:	0.998			
Model:	OLS	Adj. R-squared:	0.998			
Method:	Least Squares	F-statistic:	4.129e+05			
Date:	Thu, 23 Apr 2026	Prob (F-statistic):	0.00			
Time:	23:24:39	Log-Likelihood:	9466.6			
No. Observations:	2161	AIC:	-1.893e+04			
Df Residuals:	2157	BIC:	-1.890e+04			
Df Model:	3					
Covariance Type:	nonrobust					
=====						
	coef	std err	t	P> t	[0.025	0.975]

const	0.0008	0.001	0.932	0.352	-0.001	0.003
STLFSI	-0.0002	0.000	-0.658	0.511	-0.001	0.000
stress	0.0028	0.000	9.821	0.000	0.002	0.003
LagCorr	0.9984	0.001	1000.831	0.000	0.996	1.000
=====						
Omnibus:	1565.571	Durbin-Watson:	1.718			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	270782.406			
Skew:	2.465	Prob(JB):	0.00			
Kurtosis:	57.617	Cond. No.	33.1			

The coefficient on LagCorr is close to one and highly statistically significant, indicating that interbank connectedness is highly persistent over time. This suggests that past correlations are a dominant predictor of current correlations.

Compared to the baseline and extended models, the dynamic model provides a substantially better fit and highlights the importance of persistence in explaining interbank connectedness.

Estimation Approach

Due to violations of OLS assumptions identified earlier, all models are re-estimated using heteroskedasticity and autocorrelation consistent (HAC) standard errors to ensure robust statistical inference.

ANOVA

For model comparison, we use ANOVA based on nested OLS models estimated on a common sample. The results show that the inclusion of the lagged dependent variable significantly improves model fit ($p < 0.001$), providing strong evidence for the importance of persistence.

```
anova_lag_corrected = anova_lm(model1_common, model3_common)
print(anova_lag_corrected)
```

	df_resid	ssr	df_diff	ss_diff	F	Pr(>F)
0	2158.0	9.225716	0.0	NaN	NaN	NaN
1	2157.0	0.019824	1.0	9.205892	1.001663e+06	0.0

V. Model Validation (RMSE Comparison)

Out-of-Sample Model Comparison

To evaluate predictive performance, we implement an expanding-window time-series cross-validation approach, wherein each fold trains on past observations and tests on future data, thereby preserving the temporal structure of the dataset. This procedure is applied across three specifications: the baseline model, the extended model, and the dynamic model.

Results

The results reveal clear differences in predictive performance across models. While the baseline and extended models yield relatively high average RMSE values of approximately 0.06, the dynamic model achieves a substantially lower average RMSE of approximately 0.002 — a reduction that highlights the significant gains in out-of-sample accuracy attained by incorporating lagged interbank correlation. This marked improvement confirms that accounting for persistence in interbank connectedness is not merely a statistical refinement, but a meaningful enhancement to forecasting precision.

Interpretation

The superior performance of the dynamic model underscores the central importance of persistence in interbank connectedness. Although financial stress variables contribute meaningfully to explaining variation in correlation, lagged interbank correlation consistently emerges as the dominant predictor, a finding that holds across all cross-validation folds and confirms that the results are robust and not an artifact of any particular sample split.

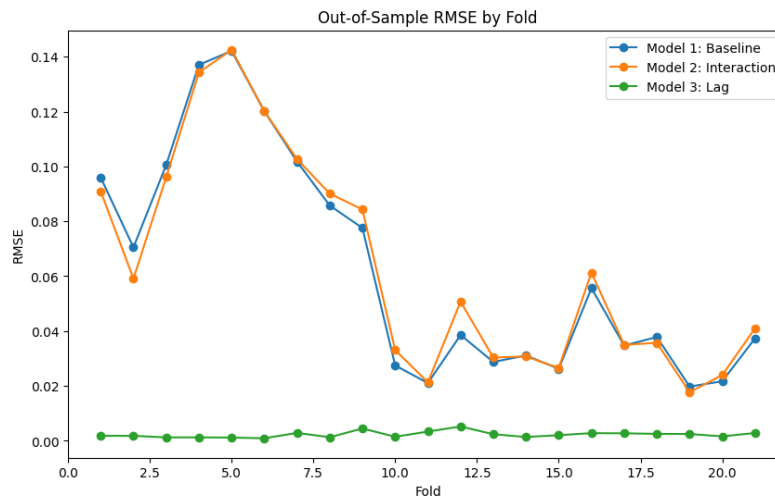
	fold	RMSE_Model1_Baseline	RMSE_Model2_Interaction	RMSE_Model3_Lag
0	1	0.095931	0.090938	0.001853
1	2	0.070516	0.059114	0.001824
2	3	0.100512	0.096093	0.001262
3	4	0.137027	0.134223	0.001264
4	5	0.142090	0.142513	0.001197
5	6	0.120154	0.120177	0.000933
6	7	0.101781	0.102656	0.002910
7	8	0.085751	0.090088	0.001307
8	9	0.077551	0.084303	0.004504
9	10	0.027488	0.033090	0.001490
10	11	0.021062	0.021332	0.003376
11	12	0.038573	0.050732	0.005237
12	13	0.028746	0.030342	0.002422
13	14	0.031113	0.030744	0.001404
14	15	0.026252	0.026547	0.002070
15	16	0.055645	0.061332	0.002797
16	17	0.034720	0.034986	0.002747
17	18	0.037853	0.035720	0.002547
18	19	0.019785	0.017712	0.002494
19	20	0.021718	0.024012	0.001630
20	21	0.037353	0.040988	0.002857

```

Average RMSE by model:
RMSE_Model1_Baseline      0.062458
RMSE_Model2_Interaction    0.063221
RMSE_Model3_Lag           0.002292
dtype: float64

```

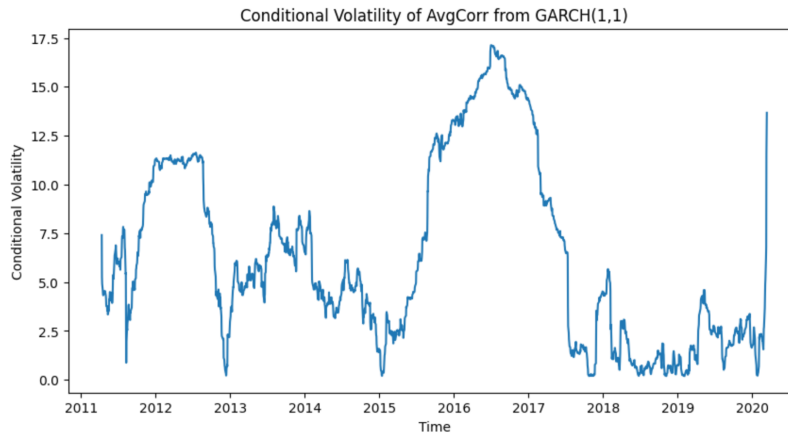
The table reports the out-of-sample RMSE for each model across cross-validation folds. The baseline and extended models exhibit relatively high prediction errors, with average RMSE values around 0.06. In contrast, the dynamic model consistently achieves much lower RMSE values, with an average of approximately 0.002. This substantial reduction in prediction error indicates that incorporating lagged interbank correlation significantly improves predictive performance.



The figure illustrates the RMSE across cross-validation folds for each model, revealing a clear and consistent performance advantage for the dynamic model. Unlike the baseline and extended models, which exhibit higher and more variable prediction errors across folds, the dynamic model maintains substantially lower RMSE throughout, demonstrating both superior accuracy and greater stability. This pattern underscores the critical role of persistence in interbank connectedness and confirms the robustness of the dynamic specification as the preferred forecasting framework.

Simple GARCH Extension

We also fitted a simple GARCH(1,1) model to AvgCorr to examine time-varying volatility in interbank connectedness. The significant ARCH (0.8715) and GARCH (0.1285) terms show that the volatility of AvgCorr is not constant over time. The conditional volatility plot further shows substantial variation, with elevated volatility around 2011–2012, the highest level during 2015–2017, and another spike near early 2020. This suggests that interbank connectedness changes dynamically across different market conditions. Therefore, the GARCH analysis serves as an exploratory extension and robustness check, providing additional support for the dynamic nature of systemic risk.



VI. Conclusion

This study investigates the relationship between financial stress and interbank connectedness among major U.S. banks. Exploratory analysis reveals that elevated financial stress is systematically associated with depressed returns, heightened volatility, and stronger interbank correlations — patterns consistent with the amplification of systemic risk during market downturns.

Regression results confirm a statistically significant positive relationship between financial stress and interbank connectedness. However, diagnostic testing exposes substantial serial correlation and heteroskedasticity in the residuals, underscoring the limitations of standard OLS estimation in this context.

Incorporating a dynamic specification that accounts for persistence yields more robust findings: lagged interbank correlation emerges as the dominant predictor of current connectedness, suggesting that systemic linkages are strongly self-reinforcing over time. Out-of-sample forecasting further validates this approach, with the dynamic model delivering substantially lower prediction errors relative to baseline specifications.

Taken together, these findings carry important implications for financial stability monitoring. Interbank connectedness exhibits a high degree of persistence, and systemic risk intensifies markedly during periods of financial stress. These results highlight the value of dynamic, persistence-aware models in capturing the evolving nature of interbank relationships and underscore the need for proactive macroprudential oversight during stress episodes.